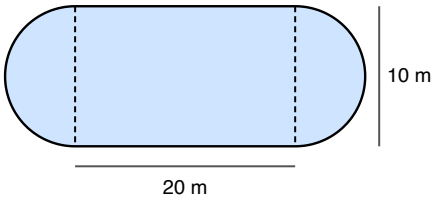
Shape and Space 7 — Circles and sectors · Solids · Similar shapes

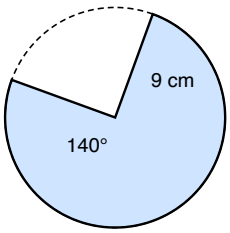
Calculator allowed. Marks in brackets. ★ = harder question. Give answers to 3 s.f. unless exact. Area of a circle πr^2, circumference $2\pi r$.

1 A running track is a rectangle 20 m by 10 m with a semicircle on each 10 m end. Find its perimeter and its area. [4]



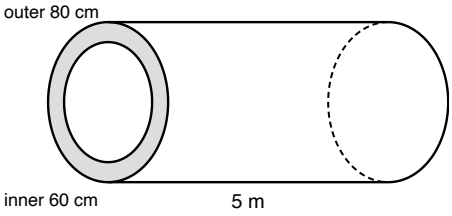
The two semicircles make one full circle of diameter 10 m (radius 5 m). Perimeter $= 2 \times 20 + \pi \times 10 = 40 + 31.4 = 71.4$ m. Area $= 20 \times 10 + \pi \times 5^2 = 200 + 78.5 = 279$ m².

2 A sector of a circle of radius 9 cm has angle 140°. Find its area and its perimeter. [4]



Area $= \dfrac{140}{360} \times \pi \times 9^2 = 99.0$ cm². Arc length $= \dfrac{140}{360} \times 2\pi \times 9 = 21.99$ cm, so perimeter $= 21.99 + 9 + 9 = 40.0$ cm (two radii as well as the arc).

3 A hollow concrete pipe is 5 m long. Its outer diameter is 80 cm and its inner diameter is 60 cm. Find the volume of concrete in m³. [5]



Work in metres: radii 0.4 m and 0.3 m. Cross-section (a ring) $= \pi(0.4^2 - 0.3^2) = \pi \times 0.07 = 0.2199$ m². Volume = cross-section × length $= 0.2199 \times 5 = 1.10$ m³.

4 Two bottles are similar. The smaller is 12 cm tall and the larger 18 cm tall. [6]

a The surface area of the smaller bottle is 150 cm². Find the surface area of the larger.

b The volume of the larger bottle is 810 cm³. Find the volume of the smaller.

Linear scale factor $k = \dfrac{18}{12} = 1.5$. **a** Areas scale by $k^2 = 2.25$: $150 \times 2.25 = 337.5$ cm². **b** Volumes scale by $k^3 = 3.375$: $810 \div 3.375 = 240$ cm³.

5 Cone A is 20 cm tall with volume 600 cm³ and curved surface area 320 cm². Cone B is similar to A and has volume 1500 cm³. [6]

a Find the height of cone B. **b** Find the curved surface area of cone B.

Volume scale factor $k^3 = \dfrac{1500}{600} = 2.5$, so $k = \sqrt[3]{2.5} = 1.3572$. **a** Height $= 20 \times 1.3572 = 27.1$ cm. **b** Area $= 320 \times k^2 = 320 \times 1.8420 = 589$ cm².

Total: 25 marks

Check: sector: $\frac{\theta}{360}$ of the area or of the circumference; perimeter adds two radii · prism/cylinder: cross-section × length · hollow shapes: subtract areas, then multiply by length · similar shapes: lengths ×k, areas ×k^2, volumes × k^3; from a volume ratio take the cube root to get k.